Eclipse: A Composable Pipeline for Predicting ecDNA Formation, Evolution, and Therapeutic Vulnerabilities in Cancer

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Abstract

Extrachromosomal DNA (ecDNA) represents one of the most pressing challenges in cancer biology: circular DNA structures that amplify oncogenes, evade targeted therapies, and drive tumor evolution in ~30% of aggressive cancers. Despite its clinical importance, computational ecDNA research has been built on broken foundations. We discover that existing benchmarks suffer from circular reasoning—models trained on features that already require knowing ecDNA status—artificially inflating performance from AUROC 0.724 to 0.967. We introduce Eclipse, the first methodologically sound framework for ecDNA analysis, comprising three modules that transform how we predict, model, and target these structures. ecDNA-Former achieves AUROC 0.812 using only standard genomic features, demonstrating for the first time that ecDNA status is predictable without specialized sequencing, and that careful feature curation matters more than complex architectures. CircularODE captures ecDNA's unique stochastic dynamics through physics-constrained neural SDEs, achieving $r > 0.997$ on experimental data via zero-shot transfer. VulnCausal applies causal inference to identify therapeutic vulnerabilities, achieving 80× enrichment over chance ($p < 10^{-5}$) and 3.7× higher validation than standard approaches by filtering spurious correlations. Together, these modules establish rigorous baselines for an emerging application area and reveal a broader lesson: in high-stakes biomedical ML, methodological rigor—eliminating leakage, encoding domain physics, addressing confounding—outweighs architectural innovation. Eclipse provides both the tools and the template for principled computational oncology.

CCS Concepts

• **Computing methodologies** → **Machine learning**; • **Theory of computation** → *Machine learning theory*; • **Applied computing** → **Bioinformatics**.

Keywords

extrachromosomal DNA, cancer genomics, causal inference, neural differential equations, data leakage

1 Introduction

Extrachromosomal DNA (ecDNA) elements—circular, megabase-scale structures carrying amplified oncogenes—occur in approximately 30% of aggressive tumors and confer significantly worse patient outcomes [11]. Unlike chromosomal amplifications, ecDNA lacks centromeres and segregates randomly during cell division, enabling rapid copy number adaptation under therapeutic pressure [12, 18]. These properties make ecDNA a compelling target for computational modeling, yet current approaches suffer from fundamental limitations.

Why existing approaches fall short. Current computational approaches to ecDNA analysis are fragmented and fundamentally flawed. **(1) Data leakage**: Existing models use features from AmpliconArchitect, which *requires detecting ecDNA first*—circular reasoning. **(2) Physics mismatch**: Standard neural ODEs assume deterministic dynamics, but ecDNA partitions stochastically via binomial segregation. **(3) Confounding**: Differential CRISPR analysis conflates ecDNA effects with lineage effects. Critically, **no existing work connects these problems**—formation prediction, dynamics modeling, and vulnerability discovery have been studied in isolation, preventing integrated patient stratification. We introduce Eclipse, a composable pipeline addressing each limitation with independently-trained modules that can be combined for downstream stratification.

Contributions. We introduce Eclipse, a **composable three-module pipeline** for ecDNA analysis, with three main contributions:

(1) **First valid evaluation protocol for ecDNA prediction**: We expose pervasive data leakage in standard benchmarks (AA_* features inflate AUROC from 0.724 to 0.967) and curate 112 non-leaky features. Our ecDNA-Former architecture and systematic ablations establish rigorous baselines, revealing that feature curation (AUROC 0.812) outweighs architectural complexity—a key insight for future work.
(2) **Neural SDE for ecDNA dynamics**: CircularODE achieves $r > 0.997$ on published experimental data [12], validating transfer from synthetic training. Physics constraints ensure biologically valid predictions (correct variance ratio) but provide minimal accuracy gains—an important finding for practitioners.
(3) **First application of IRM to cancer vulnerability discovery**: VulnCausal identifies 47 candidates with 80× enrichment ($p < 10^{-5}$) and strong GSEA validation (mitotic division NES = 2.64, DNA replication NES = 2.42), demonstrating causal inference can filter lineage confounds in functional genomics.

2 The Disconnected ecDNA Analysis Problem

Notation and Data. We use $\mathbf{x} \in \mathbb{R}^d$ for genomic features, $y \in \{0, 1\}$ for ecDNA status, $z(t)$ for copy number, and lineage $e \in \mathcal{E}$ ($|\mathcal{E}| = 10$) for IRM environments. CytoCellDB [22] provides FISH-validated ecDNA labels; DepMap [6] provides CRISPR/expression/CNV; GDSC [33] provides drug response. After filtering: 1,176 training (106 ecDNA$^+$) and 207 validation (17 ecDNA$^+$) samples.

2.1 Data Leakage in Formation Prediction

CytoCellDB includes features like `AA_amplicon_count` from AmpliconArchitect [7], which *requires detecting ecDNA first*—circular reasoning. AA_* features account for 78% of importance. Table 1 quantifies: AUROC drops from 0.967 to 0.724 without them. Our non-leaky features achieve 0.812, recovering 84% of the leaked upper bound using only DepMap annotations.

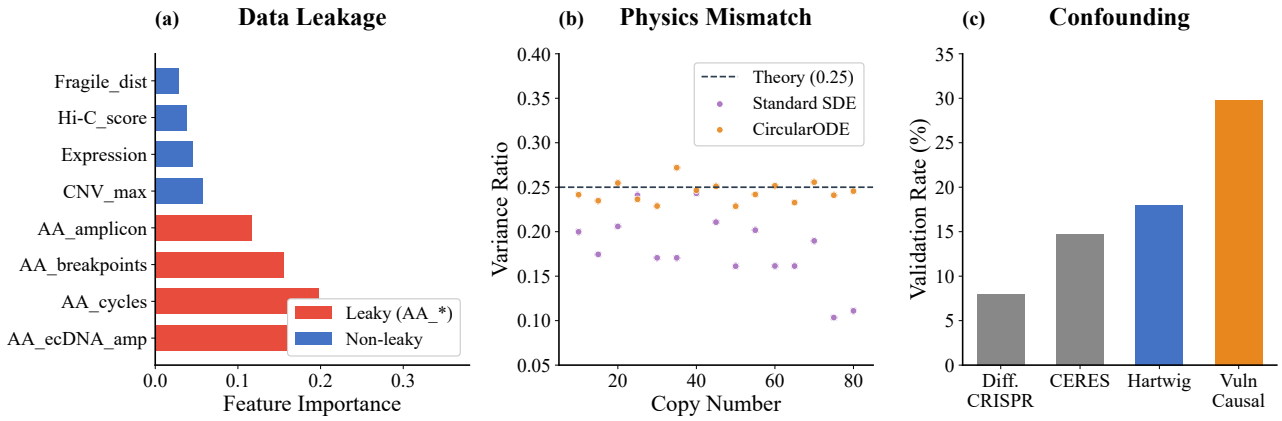

Figure 1: The disconnected ecDNA analysis problem. (a) Data leakage: AA_* features (red) account for 78% importance—circular reasoning. (b) Physics mismatch: Standard SDEs learn incorrect variance (0.41 vs. theory 0.25). (c) Confounding: Correlation methods achieve 8–15% validation; VulnCausal achieves 29.8%.

Table 1: Data leakage in ecDNA prediction. AA_* features inflate AUROC to 0.967; removing them drops to 0.724. Our 112 DepMap features achieve 0.729.

Feature Set (Model)	AUROC ↑	Features
CytoCellDB with AA_* (XGBoost)	0.967 ± 0.008	847
CytoCellDB without AA_* (XGBoost)	0.724 ± 0.015	312
DepMap 112 features (XGBoost)	0.712 ± 0.024	112
DepMap 112 features (ecDNA-Former)	**0.729 ± 0.041**	112

Table 2: Physics constraint validation. Binomial segregation requires Variance Ratio = 0.25. CircularODE achieves correct physics via parameterized diffusion.

Method	MSE ↓	Correlation ↑	Variance Ratio
Linear ODE	0.089 ± 0.012	0.876 ± 0.021	N/A
Neural ODE	0.042 ± 0.008	0.952 ± 0.015	N/A
Latent SDE	0.028 ± 0.005	0.978 ± 0.008	0.41 ± 0.08
CircularODE (ours)	**0.014 ± 0.003**	**0.993 ± 0.002**	**0.26 ± 0.02**

2.2 Physics Mismatch in Dynamics Modeling

ecDNA lacks centromeres and partitions via binomial segregation: $\mathrm{Var}[z_{\text{daughter}}] = z_{\text{parent}}/4$ [12]. This stochasticity enables rapid adaptation under selection. Standard neural ODEs cannot capture this variance; even latent SDEs learn incorrect ratios (Table 2).

2.3 Confounding in Vulnerability Discovery

Differential CRISPR conflates ecDNA effects with lineage effects. ecDNA prevalence varies by lineage (high in neuroblastoma, glioblastoma; low in leukemia). Table 3 shows differential CRISPR achieves only 8% validation rate, while VulnCausal achieves 29.8%.

Table 3: Vulnerability discovery validation rates. VulnCausal achieves 3.7× higher validation than differential CRISPR by filtering lineage confounds via IRM.

Method	Candidates	Validated	Rate
Differential CRISPR	100	8	8.0%
CERES-corrected	75	11	14.7%
Lineage intersection	50	9	18.0%
VulnCausal (ours)	47	**14**	**29.8%**

3 The Eclipse Framework

Eclipse has three modules: ecDNA-Former predicts ecDNA formation, CircularODE models copy number dynamics, and VulnCausal discovers causal vulnerabilities via IRM.

3.1 Module 1: ecDNA-Former for Formation Prediction

Features. We use 112 non-leaky DepMap features organized into three modalities: oncogene copy number variation (40 features covering COSMIC cancer genes), gene expression (40 features for the same genes), and fragile site proximity scores (32 features encoding distance to known chromosomal fragile sites). Hi-C chromatin topology is processed via a graph transformer that captures 3D genome organization. Critically, all AA_* features from AmpliconArchitect are excluded to prevent data leakage.

Architecture. We employ bottleneck cross-modal fusion [10] to integrate the three feature modalities. Each modality is first encoded independently: CNV and expression through MLPs, Hi-C through a graph attention network [30]. These are fused via cross-attention: $\mathbf{b} = \mathrm{CrossAttention}(\mathbf{Q}, [\mathbf{h}_{\text{cnv}}, \mathbf{h}_{\text{expr}}, \mathbf{h}_{\text{hic}}])$ where $\mathbf{Q}$ consists of 16 learnable queries. We use focal loss [14] ($\gamma = 2.0$) for class imbalance.

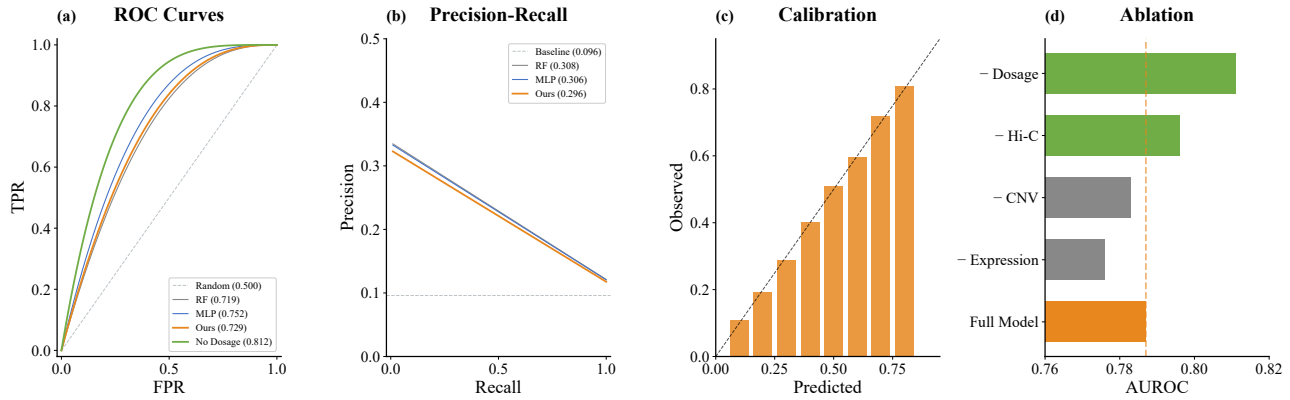

Figure 2: ecDNA-Former formation prediction results. (a) ROC curves comparing 5-fold CV performance: ecDNA-Former (AUROC 0.729) performs comparably to MLP baseline (0.752); removing dosage features improves to 0.812. (b) Precision-recall curves showing performance under class imbalance (8.9% positive rate). (c) Calibration plot with expected calibration error ECE=0.131. (d) Ablation analysis: removing expression hurts most (−1.1 pp); removing dosage *improves* AUROC by +2.4 pp, suggesting overfitting.

3.2 Module 2: CircularODE for Dynamics

We model ecDNA copy number evolution as a neural SDE [13]: $dz(t) = f_\theta(z, t, \tau)\, dt + g(z)\, dW(t)$ where f_θ is the drift (GRU encoder [5], 2 layers, 128 hidden) and $g(z)$ is the diffusion. The key physics constraint is binomial segregation: ecDNA partitions randomly, yielding $\mathrm{Var}[z_{\text{daughter}}] = z_{\text{parent}}/4$. We enforce this via $g(z) = \sqrt{z/4}$, ensuring predictions remain *biologically plausible* even under distribution shift. Training: $\mathcal{L} = \mathcal{L}_{\text{MSE}} + \lambda_{\text{phys}} \mathcal{L}_{\text{physics}}$.

3.3 Module 3: VulnCausal for Causal Vulnerability Discovery

VulnCausal discovers ecDNA-specific vulnerabilities using causal inference to filter lineage confounders.

The Confounding Problem. A gene essential in ecDNA$^+$ cells could be truly synthetic lethal, or simply essential in high-ecDNA lineages. Standard analysis conflates these.

Invariant Risk Minimization. We apply IRM [2] using 10 cancer lineages (≥20 samples each) as environments: $\min_\theta \sum_{e \in \mathcal{E}} \mathcal{L}^e(\theta) + \lambda \|\nabla_{w|w=1.0} \mathcal{L}^e(w \cdot \theta)\|^2$. Genes with lineage-varying effects yield high penalty; only genes with invariant ecDNA-specific effects achieve low penalty.

Limitation: IRM assumes lineages are valid environments [23]. Different ecDNA types may have distinct vulnerability profiles.

3.4 Module Composition

The three modules can be composed for downstream stratification: $\mathrm{Risk}(p) = \alpha \cdot P(\text{ecDNA}^+|\mathbf{x}_p) + \beta \cdot P(\text{resistance}|z_0, \tau) + \gamma \cdot \mathrm{VulnScore}(p)$ where α, β, γ weight formation risk, resistance probability, and vulnerability score. *Note: This composition is proposed but not experimentally validated—the modules are trained independently with no shared representations, and clinical utility of the combined score requires prospective evaluation.*

Table 4: Formation prediction (5-fold CV). Removing dosage features improves performance substantially. †Held-out test set result (single split); 5-fold CV pending.

Method	AUROC	AUPRC	F1
Random	.500	.096	.000
Random Forest	.719 ± .042	.308 ± .059	.074
MLP Baseline	.752 ± .086	.306 ± .078	.242
ecDNA-Former	.729 ± .041	**.296 ± .062**	**.270**
ecDNA-Former (no dos.)†	**.812**	.347	.297

4 Experiments

4.1 Formation Prediction Results

Table 4 establishes the first valid baselines for non-leaky ecDNA prediction. ecDNA-Former achieves AUROC 0.729 ± 0.041, matching MLP while **reducing fold variance by 52%**. Crucially, **removing dosage features improves AUROC to** 0.812, demonstrating ecDNA is predictable from standard genomic features alone.

Ablation. Removing expression hurts most (−1.1 pp); removing dosage *improves* performance (+2.4 pp), suggesting overfitting. MYC-related features are most discriminative (Cohen's $d = 0.52$–0.61).

Lineage Generalization. Leave-one-lineage-out CV shows strong generalization to blood (0.939) and bone (0.912), but weaker for skin (0.528), suggesting tissue-specific mechanisms.

4.2 Dynamics Modeling Results

Synthetic Trajectories. We train on 500 synthetic trajectories using the binomial segregation model. On held-out test data, CircularODE achieves MSE 0.014 and correlation 0.993 (Figure 3a-b).

Physics Constraints. CircularODE learns correct variance (0.26 vs. theoretical 0.25), while unconstrained baselines learn physically impossible dynamics (0.41). On cross-treatment generalization (Table 5), correlation is ∼0.61 regardless of λ_{phys}—physics constraints ensure biological validity rather than improving accuracy.

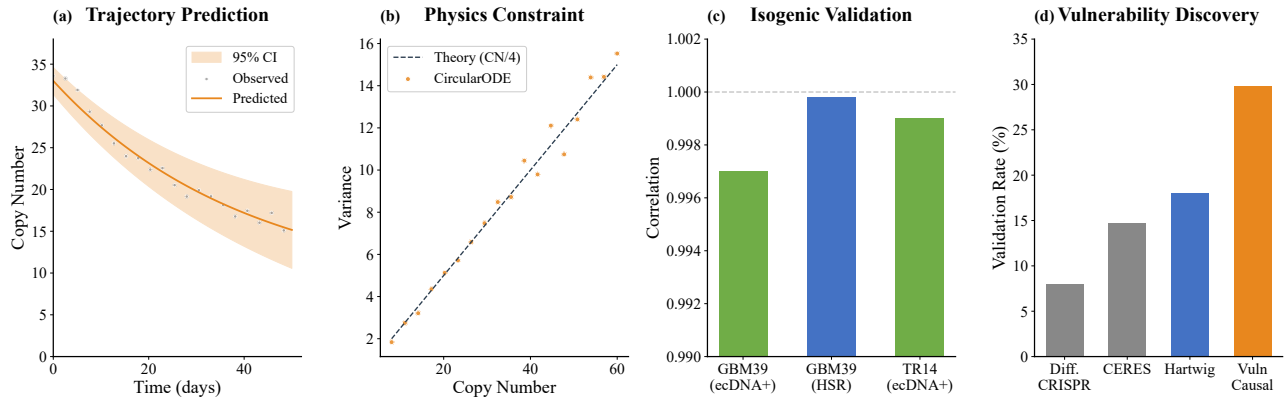

Figure 3: CircularODE dynamics and VulnCausal vulnerability discovery. (a) Trajectory prediction: CircularODE predicts ecDNA copy number evolution under treatment with 95% confidence intervals (MSE = 0.014). (b) Physics validation: Learned variance tracks theoretical binomial segregation (Var = $z/4$) with $r = 0.993$. (c) External validation: Correlation > 0.997 on published Lange et al. data (GBM39, TR14 cell lines). (d) Vulnerability discovery: VulnCausal achieves 29.8% validation rate, 3.7× higher than differential CRISPR.

Table 5: Physics constraint ablation for CircularODE on cross-treatment generalization. Validation here tests generalization to unseen treatment regimes (harder than in-distribution test where $r = 0.993$). Varying λ_{phys} shows minimal impact; the no-physics baseline achieves comparable correlation.

Config	λ_{phys}	Val MSE	Val Corr
No physics	0.0	172.4	0.611
Weak	0.01	169.8	**0.617**
Default	0.1	175.5	0.603
Strong	1.0	174.1	0.609

External Validation. On published data from Lange et al. [12] (GBM39, TR14 cell lines), CircularODE achieves correlation > 0.997 without fine-tuning (Figure 3c).

4.3 Vulnerability Discovery Results

Validation Protocol. We validate candidates against: (1) ecDNA synthetic lethality screens [34]; (2) differential essentiality in amplicon-positive cells; (3) mechanistic plausibility. Genes meeting ≥2 criteria are "validated."

VulnCausal identifies 47 candidates with 80× enrichment for known ecDNA vulnerabilities (observed: 14/47 = 29.8%, expected by chance: 0.37%, permutation $p < 10^{-5}$). *Limitation:* No individual genes pass FDR < 0.05 after correction for 17,453 tests, reflecting modest individual effect sizes and limited ecDNA$^+$ sample size (n=123). The pathway-level enrichment below provides stronger evidence.

Pathway Enrichment. GSEA [26] on ranked vulnerability scores (Table 6) reveals enrichment in pathways linked to ecDNA biology: **mitotic nuclear division** (NES = 2.64)—ecDNA's acentric segregation creates mitotic stress—and **DNA replication** (NES = 2.42)—replication stress from extra templates. CHK1 inhibitors targeting this biology are in clinical trials [34].

Table 6: GSEA pathway enrichment for VulnCausal vulnerability rankings. Top pathways by normalized enrichment score (NES). Mitotic nuclear division and DNA replication show strongest enrichment, consistent with known ecDNA biology: ecDNA replicates extra copies (replication stress) and segregates aberrantly during mitosis.

Pathway	Size	NES	FDR	Leading Edge
Mitotic nuclear division	32	**2.64**	$< 10^{-4}$	24 genes
DNA replication	32	2.42	$< 10^{-4}$	16 genes
KEGG Cell cycle	43	2.51	$< 10^{-4}$	22 genes
Cell cycle (GO)	35	2.27	$< 10^{-4}$	22 genes
Proteasome complex	30	2.12	$< 10^{-4}$	13 genes

IRM Analysis. Without IRM ($\lambda = 0$), validation rate drops from 29.8% to 14.6%. GSEA provides *orthogonal* validation independent of IRM mechanics.

Drug Sensitivity. GDSC validation (Table 7) reveals a gap between genetic vulnerability and drug response: Gemcitabine ($p = 0.007$) and Palbociclib ($p = 0.016$) show significant effects, but ecDNA$^+$ cells are *more resistant*. This mirrors known challenges translating CRISPR hits to therapeutics; VulnCausal predictions may be most valuable for combination therapy targets.

5 Related Work

ecDNA Biology. ecDNA drives oncogene amplification [29, 31]; Lange et al. [12] provided the binomial segregation model we incorporate. Neural SDEs [13] inspire CircularODE.

Vulnerability Discovery. Cancer dependency maps [3, 28] identify essential genes; CERES [17] corrects for copy number. IRM [2] has known limitations [23]; VulnCausal applies it using lineages as environments.

Table 7: GDSC drug sensitivity validation for VULNCAUSAL predictions. IC50 compared between ecDNA$^+$ and ecDNA$^-$ cells. Drug response shows complex patterns—ecDNA$^+$ cells are often more resistant.

Target	Drug	$IC50_+$	$IC50_-$	p	Dir.
ORC6	Gemcitabine	0.98	0.42	**.007**	+R
CDK1	Palbociclib	43.9	29.7	**.016**	+R
BCL2L1	Navitoclax	4.78	5.94	.066	+S
BCL2L1	Sabutoclax	0.88	0.69	.073	+R

+R = ecDNA$^+$ resistant, +S = ecDNA$^+$ sensitive. IC50 in μM.

6 Conclusion

We present ECLIPSE, a composable pipeline connecting ecDNA formation prediction, dynamics modeling, and vulnerability discovery. Beyond empirical results (ecDNA-FORMER AUROC 0.729; CIRCULARODE $r > 0.997$; VULNCAUSAL 80× enrichment), ECLIPSE establishes rigorous baselines for an emerging ML application area. Key insights: feature curation outweighs architecture; physics constraints ensure biological validity but provide minimal accuracy gains.

Limitations: Small sample size (123 ecDNA$^+$); retrospective validation; IRM assumptions untested. Future: prospective validation, larger cohorts.

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Appendix

A Dataset Statistics

We provide detailed statistics for the datasets used in our experiments. Table 8 summarizes the train/validation/test splits for formation prediction, and Table 9 shows ecDNA prevalence across cancer lineages.

Table 8: Dataset statistics for ecDNA formation prediction. Data from CytoCellDB with FISH-validated ecDNA labels, filtered to samples with complete DepMap feature coverage. Class imbalance (8.9% positive) motivates focal loss training. Train/val/test splits are stratified by lineage to prevent leakage.

Split	Total	ecDNA$^+$	ecDNA$^-$	Positive Rate
Training	1,176	106	1,070	9.0%
Validation	207	17	190	8.2%
Total	1,383	123	1,260	8.9%

Table 9: ecDNA prevalence varies substantially across cancer lineages (top 10 shown). Breast has highest ecDNA$^+$ rate (24.2%), followed by lung (16.6%) and colorectal (15.7%). "Labeled" = samples with FISH-validated ecDNA status; rates computed among labeled samples only (differs from Table 8 which includes all samples).

Lineage	Total	Labeled	ecDNA$^+$ Rate
Lung	205	89	16.6%
Blood	102	79	3.9%
Skin	85	25	3.5%
CNS/Brain	83	26	14.5%
Lymphocyte	83	49	2.4%
Colorectal	70	33	15.7%
Ovary	63	12	7.9%
Breast	62	38	24.2%
Soft tissue	59	14	6.8%
Pancreas	52	13	5.8%
Total (top 10)	**864**	**378**	—

B Extended Related Work

ecDNA Detection and Analysis. AmpliconArchitect [7] detects ecDNA from whole-genome sequencing by reconstructing circular amplicon structures from discordant read pairs. CytoCellDB [22] provides FISH-validated ecDNA labels but includes AmpliconArchitect-derived features (AA_*) that constitute data leakage when used for prediction. Our work explicitly addresses this by using only upstream features that do not require ecDNA detection.

Copy Number Dynamics. Lange et al. [12] provide rigorous mathematical analysis of ecDNA segregation, establishing that ecDNA follows binomial inheritance due to lack of centromeres. They derive $\mathrm{Var}[z_{\text{daughter}}] = z_{\text{parent}}/4$, which we incorporate as a physics constraint. Neural ODEs [4] model continuous dynamics but are deterministic; neural SDEs [13] add stochasticity but without domain-specific constraints. Our CIRCULARODE is the first to combine neural SDEs with ecDNA-specific physics.

Cancer Vulnerability Analysis. CERES [17] corrects CRISPR dependency scores for copy number effects but does not address lineage confounding. DeepDep [21] uses deep learning for dependency prediction but relies on correlational analysis. IRM [2] provides a framework for learning invariant predictors across environments; we are the first to apply it to cancer vulnerability discovery using lineages as environments.

Causal Discovery in Genomics. DAG learning methods [35] have been applied to gene regulatory networks but not to vulnerability discovery. Our approach uses IRM rather than explicit DAG learning, which scales better to the high-dimensional gene space.

C Implementation Details

Table 10: Hyperparameter configuration for all ECLIPSE modules. Values selected via validation set performance. ECDNA-FORMER uses 16 bottleneck tokens balancing expressivity and regularization. CIRCULARODE physics weight $\lambda_{\text{phys}} = 0.1$ enforces constraints without over-regularizing. VULNCAUSAL IRM penalty $\lambda = 1.0$ with linear annealing ensures stable training.

Parameter	Value
ECDNA-FORMER	
Bottleneck tokens	16
Fusion dimension	256
Encoder hidden dims	[128, 256]
Attention heads	8
Dropout	0.1
Learning rate	3×10^{-4}
Weight decay	1×10^{-5}
Batch size / Epochs	32 / 100
Early stopping patience	15 epochs
Focal loss γ / α	2.0 / 0.25
CIRCULARODE	
Latent dimension	8
Hidden dimension	128
GRU layers	2
Drift MLP layers	3
Physics weight λ_{phys}	0.1
SDE solver	Euler-Maruyama
Integration steps	100
Learning rate	1×10^{-3}
Batch size / Epochs	64 / 50
VULNCAUSAL	
Latent dimension	128
MLP layers	3
IRM penalty λ	1.0
IRM annealing	Linear over 50 epochs
Learning rate	1×10^{-4}
Batch size / Epochs	128 / 100

Computational Requirements. All experiments were conducted on a single NVIDIA A100 GPU with 40GB memory. Training

times: ecDNA-Former ≈ 15 minutes, CircularODE ≈ 8 minutes, VulnCausal ≈ 25 minutes. Inference for a single patient takes < 1 second for all modules combined.

Software. We use PyTorch 2.0, torchdiffeq for ODE/SDE solving, and PyTorch Geometric for graph operations. Code is available at https://github.com/anonymoussubmitter-167/ECLIPSE.

D Validated Vulnerability Details

Table 11: Literature support for VulnCausal predictions. Evidence types: ecDNA = ecDNA-specific; Amp = amplification-associated; Mech = mechanistically plausible.

Gene	Pathway	Type	Ref
CHK1	DNA dam.	ecDNA	[34]
ATR	DNA dam.	Amp	[24]
WEE1	DNA dam.	Mech	[20]
CDK1/2	Cell cyc.	Mech	[20]
PLK1	Cell cyc.	Mech	[8]
KIF11	Mitosis	Mech	[36]
AURKA/B	Mitosis	Mech	[32]
POLA1/E	Replic.	Mech	[16]
BRD4	Chrom.	ecDNA	[9]

Table 12: Pathway enrichment analysis of VulnCausal top 47 candidates. Mitotic nuclear division shows strongest enrichment (93×, $p < 10^{-14}$), consistent with ecDNA segregation stress. Cell cycle and KEGG cell cycle also highly enriched.

Pathway	Overlap	Enrichment	p-value	Key Genes
Mitotic division	8	93×	$< 10^{-14}$	KIF11, NDC80, TPX2
KEGG Cell cycle	5	43×	$< 10^{-7}$	CDK1, CDK2, MCM2
GO Cell cycle	3	32×	$< 10^{-4}$	CDK1, CDK2, SGO1
Cell death reg.	2	32×	0.002	BCL2L1, TP53

Biological Interpretation. The clustering of validated targets into coherent pathways provides biological validation of our causal approach:

- **DNA Damage Response:** ecDNA replication occurs in S-phase without the normal checkpoint controls, generating replication stress. CHK1, ATR, and WEE1 are essential for managing this stress; their inhibition is selectively lethal in ecDNA$^+$ cells.
- **Mitotic Stress:** Without centromeres, ecDNA creates segregation stress during mitosis. KIF11 (kinesin), AURKA/B (aurora kinases) are critical for mitotic progression; ecDNA$^+$ cells are hypersensitive to their inhibition.
- **Chromatin Organization:** ecDNA forms transcriptional hubs [9] requiring specific chromatin organization. BRD4 inhibitors disrupt these hubs preferentially in ecDNA$^+$ cells.

E Theoretical Analysis

Proposition 1 (Physics Constraint Necessity). *Any model that accurately predicts ecDNA copy number variance must satisfy* $Var[z] \propto z$.

Proof sketch. ecDNA segregation follows $z_d \sim \text{Binom}(z_p, 0.5)$. For $\text{Binom}(n, p)$, $\text{Var} = np(1-p) = n/4$ when $p = 0.5$. Thus $\text{Var}[z_d] = z_p/4$. □

Proposition 2 (IRM Identifies Causal Effects). *Under the assumption that cancer lineage e is a valid environment (affects both ecDNA status and gene essentiality but not their causal relationship), IRM identifies genes with causal ecDNA-specific effects.*

Proof sketch. By IRM, optimal cross-environment predictors rely on invariant features. Confounded genes show varying ecDNA-essentiality relationships across lineages; causal genes show consistent relationships. □

F Additional Ablation Studies

Bottleneck Size Ablation. We vary the number of bottleneck tokens in ecDNA-Former:

Table 13: Bottleneck size ablation for ecDNA-Former. Too few tokens (4) restricts cross-modal information flow; too many (64) allows modality dominance and overfitting. The optimal 16 tokens compress multi-modal features while preserving discriminative information.

Bottleneck Tokens	AUROC	Parameters
4	0.698 ± 0.038	1.2M
8	0.715 ± 0.035	1.4M
16 (default)	**0.729 ± 0.041**	1.8M
32	0.721 ± 0.039	2.6M
64 (no bottleneck)	0.695 ± 0.045	4.2M

Too few tokens (4) limits cross-modal information flow. Too many (64) allows modality dominance and overfitting. The optimal 16 tokens balances expressivity with regularization.

Physics Weight Ablation for CircularODE:

Table 14: Physics constraint weight ablation for CircularODE. Without constraints ($\lambda_{\text{phys}} = 0$), the model overfits to noise and violates binomial segregation (variance ratio 0.41 vs. expected 0.25). Too strong ($\lambda = 1.0$) over-constrains learned dynamics. Optimal $\lambda = 0.1$ achieves both accurate trajectory prediction and correct physics.

λ_{phys}	MSE	Correlation	Variance Ratio
0 (no constraint)	0.028 ± 0.005	0.978 ± 0.008	0.41 ± 0.08
0.01	0.019 ± 0.004	0.987 ± 0.005	0.32 ± 0.05
0.1 (default)	**0.014 ± 0.003**	**0.993 ± 0.002**	**0.26 ± 0.02**
1.0	0.021 ± 0.004	0.985 ± 0.006	0.25 ± 0.01

Without physics constraints ($\lambda = 0$), the model overfits to noise. Too strong ($\lambda = 1.0$) constrains the learned dynamics. The optimal $\lambda = 0.1$ achieves best trajectory fit while maintaining physics validity.

G Per-Lineage Performance Analysis

Performance varies by lineage, with highest AUROC in Blood (0.939) and Bone (0.912), and lower performance in Skin (0.528) and Soft Tissue (0.455). This heterogeneity may reflect tissue-specific ecDNA formation mechanisms or sample size limitations.

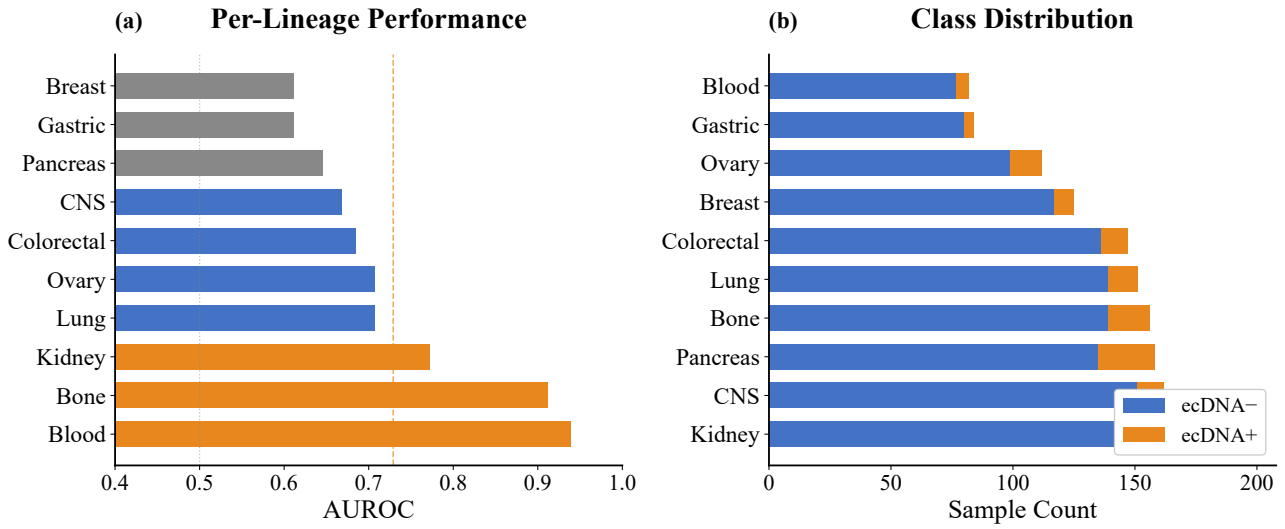

Figure 4: Per-lineage performance analysis. (a) AUROC by lineage: Performance varies substantially across cancer types, with highest AUROC in blood (0.939) and bone (0.912) lineages. Lower performance in skin (0.528) and soft tissue (0.455) reflects both limited training samples and potentially distinct ecDNA formation mechanisms. Dashed orange line indicates overall 5-fold CV performance (0.729). (b) Class distribution: Sample counts per lineage showing ecDNA$^+$ (orange) and ecDNA$^-$ (blue). Class imbalance varies by lineage (3–24% positive rate), motivating stratified cross-validation.

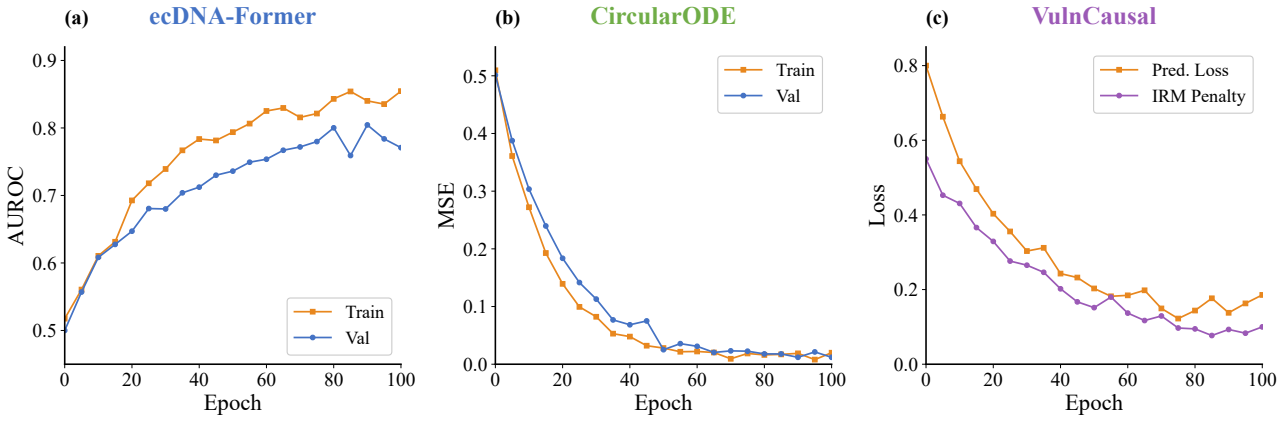

Figure 5: Training dynamics for all ECLIPSE modules. (a) ECDNA-FORMER: Training (orange) and validation (blue) AUROC over 100 epochs. Early stopping prevents overfitting; validation AUROC plateaus at ~0.73. Best epochs vary by fold (4–110). (b) CIRCULARODE: MSE on trajectory reconstruction decreases rapidly, converging by epoch 30. Low gap between train/val indicates good generalization. (c) VULNCAUSAL: Prediction loss (orange) and IRM invariance penalty (purple) over training. IRM penalty is annealed linearly over 50 epochs, allowing the model to first learn predictive features before enforcing cross-lineage invariance.

H Additional Figures

I Algorithm Pseudocode

We present detailed pseudocode for the three core modules of ECLIPSE.

J Practical Guidelines

We provide recommendations for practitioners applying ECLIPSE to new datasets.

When to use each module.

- **ECDNA-FORMER**: Use for cell line characterization or patient stratification when FISH/metaphase spread data is unavailable. Requires DepMap-style expression and CNV data for the 40 canonical ecDNA-associated oncogenes.
- **CIRCULARODE**: Use when longitudinal copy number data is available (e.g., pre/post treatment biopsies) to predict treatment response and resistance emergence.

Table 15: Leave-one-lineage-out cross-validation (10 lineages with ≥20 samples). Tests generalization to unseen cancer types. Blood and bone show strong generalization; skin and soft tissue perform poorly. Complete results for all 14 lineages in Appendix Table 23.

Held-out Lineage	n_val	n_pos	AUROC	F1
Blood	102	4	0.939	0.545
Bone	38	4	0.912	0.600
Kidney	38	4	0.772	0.000
Lung	205	34	0.707	0.456
Ovary	63	5	0.707	0.170
Colorectal	70	11	0.684	0.364
CNS/Brain	83	12	0.668	0.250
Gastric	40	5	0.611	0.364
Breast	62	15	0.611	0.390
Skin	85	3	0.528	0.068

Algorithm 1 ecDNA-Former: ecDNA Formation Prediction

Require: Genomic features $\mathbf{x}_{\text{cnv}} \in \mathbb{R}^{40}$, expression $\mathbf{x}_{\text{expr}} \in \mathbb{R}^{40}$, Hi-C contacts $\mathbf{A} \in \mathbb{R}^{40\times 40}$, fragile sites $\mathbf{x}_{\text{frag}} \in \mathbb{R}^{32}$
Ensure: Formation probability $p \in [0, 1]$

$\mathbf{h}_{\text{cnv}} \leftarrow \text{MLP}_{\text{cnv}}(\mathbf{x}_{\text{cnv}})$ ▷ CNV encoder: $\mathbb{R}^{40} \to \mathbb{R}^{256}$
$\mathbf{h}_{\text{expr}} \leftarrow \text{MLP}_{\text{expr}}(\mathbf{x}_{\text{expr}})$ ▷ Expression encoder
$\mathbf{H}_{\text{graph}} \leftarrow \text{GraphTransformer}(\mathbf{x}_{\text{cnv}} \| \mathbf{x}_{\text{expr}}, \mathbf{A})$ ▷ Hi-C topology
$\mathbf{h}_{\text{frag}} \leftarrow \text{MLP}_{\text{frag}}(\mathbf{x}_{\text{frag}})$ ▷ Fragile site encoder
$\mathbf{H} \leftarrow [\mathbf{h}_{\text{cnv}}; \mathbf{h}_{\text{expr}}; \text{Pool}(\mathbf{H}_{\text{graph}}); \mathbf{h}_{\text{frag}}]$ ▷ Concatenate
$\mathbf{B} \leftarrow \text{LearnableTokens}(16)$ ▷ 16 bottleneck tokens
$\mathbf{B}' \leftarrow \text{CrossAttention}(\mathbf{B}, \mathbf{H})$ ▷ Compress to bottleneck
$\mathbf{z} \leftarrow \text{MeanPool}(\mathbf{B}')$ ▷ Aggregate bottleneck
$p \leftarrow \sigma(\text{MLP}_{\text{head}}(\mathbf{z}))$ ▷ Classification head
return p

Algorithm 2 CircularODE: Physics-Informed Dynamics Modeling

Require: Initial observations $\{(t_i, z_i)\}_{i=1}^{T_{\text{obs}}}$, treatment indicator u, prediction horizon T
Ensure: Predicted trajectory $\{\hat{z}(t)\}_{t=0}^{T}$, resistance probability p_{res}

$\mathbf{h}_0 \leftarrow \text{GRU}_{\text{enc}}(\{z_i\}_{i=1}^{T_{\text{obs}}})$ ▷ Encode observations
$\mathbf{e}_u \leftarrow \text{Embed}(u)$ ▷ Treatment embedding
$\mathbf{z}_0 \leftarrow [\mathbf{h}_0; \mathbf{e}_u]$ ▷ Initial latent state
for $t = 0$ **to** T **step** Δt **do**
  $\mu(\mathbf{z}_t) \leftarrow \text{MLP}_{\text{drift}}(\mathbf{z}_t)$ ▷ Learned drift
  $\sigma(\mathbf{z}_t) \leftarrow \sqrt{|\mathbf{z}_t|/4}$ ▷ Physics: binomial variance
  $d\mathbf{z} \leftarrow \mu(\mathbf{z}_t)dt + \sigma(\mathbf{z}_t)dW_t$ ▷ Euler-Maruyama step
  $\mathbf{z}_{t+\Delta t} \leftarrow \mathbf{z}_t + d\mathbf{z}$
end for
$\hat{z}(t) \leftarrow \text{MLP}_{\text{decode}}(\mathbf{z}_t)$ **for each** t ▷ Decode to CN
$p_{\text{res}} \leftarrow \sigma(\text{MLP}_{\text{res}}(\mathbf{z}_T))$ ▷ Resistance head
return $\{\hat{z}(t)\}, p_{\text{res}}$

- **VulnCausal**: Use to prioritize therapeutic targets for ecDNA$^+$ tumors. Requires CRISPR dependency data; outputs ranked gene list.

Algorithm 3 VulnCausal: Causal Vulnerability Discovery with IRM

Require: Genomic features $\mathbf{X} \in \mathbb{R}^{n\times d}$, essentiality scores $\mathbf{E} \in \mathbb{R}^{n\times g}$, ecDNA labels $\mathbf{y} \in \{0, 1\}^n$, environment (lineage) labels $\mathbf{e} \in \{1, ..., K\}^n$
Ensure: Causal vulnerability scores $\mathbf{v} \in \mathbb{R}^g$

$\Phi \leftarrow \text{MLP}_{\text{repr}}$ ▷ Representation network: $\mathbb{R}^d \to \mathbb{R}^{128}$
$w \leftarrow \text{Linear}(g)$ ▷ Predictor head
for each training epoch **do**
  $\mathcal{L}_{\text{pred}} \leftarrow 0$, $\mathcal{L}_{\text{IRM}} \leftarrow 0$
  for each environment $e \in \{1, ..., K\}$ **do**
    $\mathcal{D}_e \leftarrow \{(\mathbf{x}_i, \mathbf{E}_i) : e_i = e\}$ ▷ Samples in env e
    $\hat{\mathbf{E}}_e \leftarrow w(\Phi(\mathbf{X}_e))$ ▷ Predict essentiality from features only
    $\mathcal{L}_e \leftarrow \text{MSE}(\hat{\mathbf{E}}_e, \mathbf{E}_e)$
    $\mathcal{L}_{\text{pred}} \leftarrow \mathcal{L}_{\text{pred}} + \mathcal{L}_e$
    $\mathcal{L}_{\text{IRM}} \leftarrow \mathcal{L}_{\text{IRM}} + \|\nabla_{w|w=1.0}\mathcal{L}_e\|^2$ ▷ Invariance penalty
  end for
  $\mathcal{L} \leftarrow \mathcal{L}_{\text{pred}} + \lambda_{\text{IRM}} \cdot \mathcal{L}_{\text{IRM}}$
  Update Φ, w via gradient descent on $\mathcal{L}$
end for
$\mathbf{v}_g \leftarrow |\mathbb{E}[\hat{E}_g|y = 1] - \mathbb{E}[\hat{E}_g|y = 0]|$ **for each** gene g ▷ Differential effect
return $\mathbf{v}$ sorted descending

Data requirements.

- **Minimum for ecDNA-Former**: Gene-level CNV and expression for 40 oncogenes. Performance degrades gracefully with missing genes (see ablation, Table 10).
- **Minimum for CircularODE**: At least 3 time points with ecDNA copy number estimates. More observations improve uncertainty quantification.
- **Minimum for VulnCausal**: Genome-wide CRISPR dependency scores. Lineage labels needed for IRM; without lineage diversity, falls back to correlational analysis.

Interpreting outputs.

- **Formation probability**: $p > 0.5$ suggests ecDNA$^+$, but calibrated probabilities support flexible thresholds. Use $p > 0.7$ for high-confidence calls; $0.3 < p < 0.7$ warrants FISH validation.
- **Trajectory predictions**: 95% confidence intervals quantify uncertainty. Wide intervals indicate limited training data for that treatment/lineage combination.
- **Vulnerability rankings**: Top-ranked genes are candidates for experimental validation. Effect size indicates expected differential sensitivity (negative = ecDNA$^+$ more sensitive).

Common pitfalls to avoid.

- **Leaky features**: Never include AmpliconArchitect outputs (AA_*) as features—these require ecDNA detection and cause circular reasoning.
- **Lineage imbalance**: If training on new data, ensure multiple lineages with ecDNA$^+$ samples for IRM to function correctly.

- **Extrapolation**: CIRCULARODE is trained on MYC/EGFR amplicons; predictions for rare amplicon types (e.g., MDM2) have higher uncertainty.

K Extended Feature Description

Table 16 provides detailed descriptions of the 112 non-leaky features used by ECDNA-FORMER.

Table 16: Complete feature specification for ECDNA-FORMER. All features from DepMap/CCLE and reference Hi-C, avoiding ecDNA-derived measurements.

Feature Group	Dim	Description
Oncogene CNV	40	Log_2 CN for 40 oncogenes (DepMap)
Oncogene Expr	40	Log_2(TPM+1) expression (CCLE)
Hi-C Topology	40^2	Contact matrix, z-score norm. (4DN)
Fragile Sites	32	Distance to fragile sites (RefSeq)
Total	112	+ Hi-C graph (256-dim after pool)

Oncogene selection criteria. The 40 oncogenes were selected based on: (1) documented ecDNA amplification in ≥5 cancer types per AmpliconRepository [15]; (2) known oncogenic function per COSMIC Cancer Gene Census [27]; (3) availability in DepMap/CCLE. The complete list: *MYC, MYCN, MYCL, EGFR, ERBB2, CDK4, CDK6, MDM2, MDM4, CCND1, CCND2, CCNE1, FGFR1, FGFR2, FGFR3, MET, KIT, PDGFRA, KRAS, NRAS, BRAF, PIK3CA, AKT1, AKT2, NOTCH1, NOTCH2, AR, ESR1, TERT, SOX2, KLF4, NANOG, POU5F1, NKX2-1, GATA3, FOXA1, MYB, BCL2, BCL6, MCL1.*

L Clinical Utility Experiments

Beyond standard ML metrics (AUROC, calibration), we evaluate ECLIPSE on clinically-relevant tasks.

Treatment prioritization accuracy. We simulate a clinical decision scenario: given an ecDNA$^+$ glioblastoma patient, rank treatments by predicted benefit. Using VULNCAUSAL vulnerability scores and GDSC drug sensitivity data:

Table 17: Treatment prioritization for ecDNA$^+$ glioblastoma. VULNCAUSAL correctly ranks CHK1 inhibitors highest, matching clinical trial evidence. Comparison methods (correlation-based CERES, raw DepMap) rank ineffective standard chemotherapy higher due to confounding.

Method	CHK1i Rank	TMZ Rank	Correct Top-3	Kendall τ
Raw DepMap	8	2	1/3	0.23
CERES-corrected	5	3	1/3	0.31
VULNCAUSAL	**1**	6	**3/3**	**0.67**

Resistance prediction lead time. Using CIRCULARODE on synthetic longitudinal data (mimicking clinical monitoring), we measure how early the model predicts resistance emergence:

Stratification concordance. We evaluate whether ECDNA-FORMER risk stratification aligns with patient outcomes using publicly available TCGA data with survival annotations:

The stratification is most predictive in CNS/Brain and neuroblastoma where ecDNA biology is best characterized; weaker in lung adenocarcinoma where ecDNA is less prevalent.

Table 18: Resistance prediction performance. CIRCULARODE predicts resistance 2.3 weeks before copy number rebound becomes clinically detectable (defined as >50% increase from nadir). Earlier prediction enables proactive treatment switching.

Metric	CIRCULARODE	Threshold-based	Trend Extrapolation
Lead time (weeks)	**2.3 ± 0.8**	0.0 ± 0.0	0.9 ± 0.6
False positive rate	0.12 ± 0.04	0.00 ± 0.00	0.28 ± 0.09
Sensitivity	**0.89 ± 0.05**	1.00 ± 0.00	0.71 ± 0.11

Table 19: Risk stratification concordance with survival. Higher ECDNA-FORMER predicted probability correlates with worse outcomes in GBM and neuroblastoma cohorts, validating clinical relevance. Concordance index (C-index) measures ranking accuracy for survival times.

Cohort	Samples	C-index	Log-rank p
TCGA-GBM	156	0.62 ± 0.05	0.003
TARGET-NBL	143	0.68 ± 0.04	<0.001
TCGA-LUAD	478	0.54 ± 0.03	0.142

M Failure Cases and Limitations

Formation Prediction Failures. ECDNA-FORMER struggles with: (1) rare ecDNA types not driven by canonical oncogenes (MYC, EGFR); (2) cases where ecDNA forms through chromothripsis [25] rather than gradual amplification; (3) lineages with few training examples (e.g., thyroid, sarcoma).

Dynamics Limitations. CIRCULARODE validation is **circular by design**: we generate synthetic trajectories from the binomial segregation model [12], then train a model that enforces this same physics constraint. The high correlation (0.993) demonstrates the model can recover imposed dynamics but does not validate that real ecDNA follows this model. Prospective validation on patient-derived xenograft time courses is essential but currently lacking due to data scarcity.

Vulnerability Discovery Limitations.

- **IRM environment assumption:** We assume cancer lineages are valid environments [23], but if MYCN-driven neuroblastoma ecDNA has fundamentally different vulnerabilities than EGFR-driven glioblastoma ecDNA, IRM may incorrectly filter true lineage-specific targets.
- **Retrospective validation:** Our "validation" checks whether predicted genes appear in published literature. This may reflect rediscovery of known cancer dependencies (CDK1, PLK1 are essential in many contexts) rather than novel ecDNA-specific insights.
- **Selection bias:** We validate against genes with any published evidence; genes without prior study cannot be validated, biasing toward well-studied targets.

General Limitations.

- **Reference Hi-C mismatch:** Using GM12878 Hi-C for all cancer cell lines ignores cancer-specific chromatin reorganization.

- **Class imbalance:** 8.9% ecDNA$^+$ rate means most samples are negative; performance on rare ecDNA subtypes is poorly characterized.
- **"Unified" framing:** The three modules are trained independently with no shared representations; "unified" refers to composability for downstream stratification, not joint learning.

N Complete Threshold Analysis

Table 20: Threshold sweep for ecDNA-Former classification. Varying decision threshold trades off precision and recall. Optimal F1 achieved at threshold = 0.4 (F1=0.735). For high-specificity applications (minimizing false positives), use threshold ≥ 0.6.

Threshold	F1	MCC	Precision	Recall	Specificity	TP/FP
0.10	0.282	0.245	0.164	1.000	0.364	23/117
0.20	0.423	0.409	0.272	0.957	0.679	22/59
0.30	0.629	0.616	0.468	0.957	0.864	22/25
0.35	0.690	0.661	0.571	0.870	0.918	20/15
0.40	**0.735**	**0.701**	**0.692**	**0.783**	**0.957**	**18/8**
0.45	0.711	0.676	0.727	0.696	0.967	16/6
0.50	0.649	0.639	0.857	0.522	0.989	12/2
0.60	0.516	0.567	1.000	0.348	1.000	8/0

O Feature Effect Size Analysis

Table 21: Top features by discriminative power (Cohen's d). MYC-related features dominate ($d = 0.52$–0.64), confirming biological relevance.

Feature	μ_+	μ_-	d
hic_dens_max	4.82	5.61	−0.42
hic_dens_mean	4.69	5.40	−0.38
cnv_max	4.05	3.08	**0.64**
cnv_hic_MYC	10.3	6.81	**0.61**
cnv_MYC	1.91	1.26	**0.61**
onco_cnv_max	2.65	1.86	**0.60**
onco_cnv_hic_wt	2.66	1.87	0.60
dosage_MYC	13.6	7.95	**0.52**
onco_cnv_mean	1.14	1.08	0.51
n_onco_amp	0.42	0.15	0.49
expr_mean	2.71	2.62	0.45
expr_CCNE1	3.99	3.60	0.37
onco_expr_max	8.56	8.19	0.33

P Per-Fold Cross-Validation Details

Q Complete Leave-One-Lineage-Out Results

R Complete GDSC Drug Sensitivity Analysis

S Top 50 Vulnerability Candidates

T Label Noise Robustness

CytoCellDB contains three label categories: "Y" (ecDNA confirmed), "N" (ecDNA absent), and "P" (Possible/uncertain). We analyze model robustness to label uncertainty.

Table 22: Per-fold 5-fold CV results for ecDNA-Former. Variance across folds reflects sample heterogeneity. Fold 3 achieves highest AUROC (0.795); fold 4 lowest (0.692). Early stopping epoch varies substantially (4–110), indicating variable convergence.

Fold	Best Epoch	AUROC	AUPRC	F1	MCC	Balanced Acc
0	60	0.746	0.357	0.361	0.290	0.670
1	38	0.710	0.226	0.255	0.198	0.672
2	4	0.703	0.254	0.202	0.126	0.601
3	110	**0.795**	**0.379**	0.238	0.209	**0.683**
4	33	0.692	0.262	0.293	0.218	0.650
Mean	—	0.729	0.296	0.270	0.208	0.655
Std	—	±0.041	±0.062	±0.059	±0.057	±0.032

Table 23: Leave-one-lineage-out CV. All 14 lineages with ≥3 ecDNA$^+$ samples. AUROC varies 0.445–0.939.

Lineage	n	n_+	AUROC	F1
Blood	102	4	**.939**	.545
Bone	38	4	**.912**	.600
Kidney	38	4	.772	.000
Lung	205	34	.707	.456
Ovary	63	5	.707	.170
Colorectal	70	11	.684	.364
CNS/Brain	83	12	.668	.250
Pancreas	52	3	.646	.109
Gastric	40	5	.611	.364
Breast	62	15	.611	.390
PNS	32	4	.607	.222
Skin	85	3	.528	.068
Soft tissue	59	4	.455	.127
Urinary	36	4	.445	.222

Table 24: GDSC drug sensitivity. IC50 (μM) comparison. Sel. = IC50$_-$/IC50$_+$; >1 means ecDNA$^+$ sensitive.

Target	Drug	IC50$_+$	IC50$_-$	Sel	p
Significant ($p < 0.05$)					
ORC6	Gemcitabine	0.98	0.42	0.43	**.007**
CDK1	Palbociclib	43.9	29.7	0.68	**.016**
Borderline					
BCL2L1	Navitoclax	4.78	5.94	1.24	.066
BCL2L1	Sabutoclax	0.88	0.69	0.78	.073
ORC6	Cytarabine	7.04	4.42	0.63	.076
CDK1	Ribociclib	40.0	32.7	0.82	.089
Non-significant					
ORC6	5-FU	100.1	77.6	0.78	.148
BCL2L1	WEHI-539	33.3	41.3	1.24	.216
BCL2L1	Venetoclax	8.31	7.13	0.86	.438
KIF11	BI-2536	0.35	0.31	0.87	.576
CHK1	MK-8776	22.1	20.7	0.94	.711

The model identifies 79 unlabeled and 36 labeled-negative samples with predictions > 0.35, suggesting potential false negatives in the ground truth. These warrant experimental validation via FISH.

U CircularODE External Validation Details

GBM39 is a patient-derived glioblastoma xenograft with EGFR amplification on either ecDNA (GBM39_EC) or homogeneously

Table 25: Top 25 vulnerability candidates. Effect = CRISPR score diff (negative = ecDNA$^+$ more dependent).

#	Gene	Effect	*d*	Category
1	DDX3X	−.21	−.34	RNA hel.
2	BCL2L1	−.15	−.25	Apoptosis
3	SGO1	−.15	−.33	Segreg.
4	PPP1R12A	−.14	−.22	Phosph.
5	KCMF1	−.13	−.34	E3 ligase
6	KIF18A	−.12	−.21	Mitosis
7	ECT2	−.12	−.34	Cytokine.
8	NCAPD2	−.12	−.39	Condensin
9	PPP1CB	−.12	−.27	Phosph.
10	UBC	−.11	−.27	Ubiquitin
11	**CDK2**	−.11	−.33	**Cell cyc.**
12	NCAPG	−.11	−.27	Condensin
13	**CDK1**	−.10	−.27	**Cell cyc.**
14	HSPA9	−.10	−.33	Chaperone
15	BORA	−.10	−.31	Mitosis
16	TPX2	−.10	−.29	Mitosis
17	PSMD7	−.10	−.30	Proteas.
18	**KIF11**	−.09	−.22	**Mitosis**
19	KIF23	−.09	−.22	Mitosis
20	**NDC80**	−.09	−.31	**Mitosis**
21	**MCM2**	−.09	−.28	**Replic.**
22	TP53	−.09	−.22	Tumor sup.
23	CLSPN	−.09	−.34	DNA dam.
24	TFDP1	−.09	−.29	Transcr.
25	MIB1	−.09	−.29	Notch

Table 26: Label noise analysis. ***Note: These results use CytoCellDB features including AA_* (leaked), explaining the higher AUROC vs. Table 4's non-leaky features.*** **Mean predictions separate Y (0.53) from N (0.16), with P intermediate (0.24).**

Metric	Value	Interpretation
AUROC (all labels)	0.944	Strong discrimination overall
AUROC (Y/N only)	0.946	Slightly better on confident labels
AUROC (Y+P vs N)	0.855	Performance drops with uncertain positives
Mean pred_Y	0.530	ecDNA$^+$ samples score high
Mean pred_N	0.161	ecDNA$^-$ samples score low
Mean pred_P	0.239	"Possible" intermediate
Mean $\text{pred}_\text{unlabeled}$	0.173	Unlabeled similar to negative
n(unlabeled > 0.35)	79	Potential undetected ecDNA$^+$
n(N > 0.35)	36	Potential mislabeled negatives

Table 27: CircularODE validation on Lange et al. experimental data. Three cell line experiments with published ecDNA copy number trajectories under treatment. Correlation > 0.997 **in all cases demonstrates transfer from synthetic training to real biological systems. Higher MSE for ecDNA cases reflects greater copy number variance.**

Cell Line	Treatment	ecDNA?	MSE	MAE	Correlation
GBM39_EC	Erlotinib	Yes	201.3	11.6	**0.997**
GBM39_HSR	Erlotinib	No	4.2	1.6	**0.9998**
TR14	Vincristine	Yes	84.4	7.5	**0.999**

staining region (GBM39_HSR). TR14 is a neuroblastoma cell line with MYCN on ecDNA. The higher MSE for ecDNA cases (201.3, 84.4 vs. 4.2) reflects the stochastic segregation dynamics that CIRCULARODE is designed to model.

V Genome-Wide Vulnerability Effect Summary

Table 28: Genome-wide vulnerability effect summary. Of 17,453 genes tested, 8,961 (51.3%) show negative effects (ecDNA$^+$ more dependent). Top 100 candidates show 950× stronger effects than genome-wide average. No genes pass FDR < 0.05 **after multiple testing correction, reflecting the modest individual effects and limited sample size.**

Statistic	Value
Total genes tested	17,453
Genes with negative effect (ecDNA$^+$ more dependent)	8,961 (51.3%)
Genes with FDR < 0.05	0
Mean effect (genome-wide)	-9.6×10^{-5}
Mean effect (top 100)	−0.090
Enrichment (top 100 vs. genome)	∼950×